

## **PROJECT PLANNING AND CONTROL**

### **MFET 3008/MFET 5040**

#### **Final Case Study**

##### **Instruction for your submission**

1. Please read carefully all instructions before you prepare your answer.
2. This project should be done in a group.
3. Type your report using Times New Roman, Arial or similar types of fonts with size 12, or write it clearly by hand on one side (not both sides) of A4 sized paper. The templates for the answer using MSExcel are provided by the lecturer or downloaded from the course homepage.
4. Each group must submit the following two items
  - a. A report containing the detailed calculation procedure: It should include clear and detailed explanation of the logic that you have used to obtain your answer. Clearly state all assumptions that have been made. Use the cover sheet provided in the study guide. Without it=5% penalty.
  - b. A diskette or a CD containing all Gantt charts and other necessary document: Note that this diskette/CD will be used only for the convenience of the marking process. So all answers in your CD or Diskette should be printed out and submitted for the marking. Remember! Only the print-out copy of your report will be used for the marking.
5. The submitted answers **will not be returned** after the marking. However, the feedback report will be provided. The marking will be made using the mark sheet attached at the end of this instruction.
6. The instructor's answer for this assignment will not be provided because there would be several different answers depending on the assumptions that each answer used. So it is very important that each decision made to obtain your answer should be justified. This is similar to what a project manager will face in a company in the future, i.e., to solve a certain problem in an organization, there would be several alternative solutions. The project manager should choose one best option through rigorous justifiable decision making processes.
7. Due date will be announced by the lecturer during the lecture.

**No late submission will be accepted.**

## Summary:

Your company has been granted a contract to design, fabricate and commission a new computer-controlled parcel transport, storage and database system, for automatic placement, storage and retrieval of standardised shipping containers. The system uses an overhead conveyor track system, conveyor-robot transporter units, racks with standard size shipping containers and storage buckets, and a computerized database for automatic placement and retrieval of parcels and record keeping. The main tasks of this project have been broken down and listed in Table 1.

**Q1. WBS, OBS and LRC (total 10 marks):** Based on the tasks and required staffs for each task listed in Table 1,

- a. Create an Organizational Breakdown Structure (OBS) and a Work Breakdown Structure (WBS) for the project including a standard coding system. Both structures should have at least four levels of hierarchy (4 marks).
- b. Create a Linear Responsibility Chart (LRC) to allocate the tasks into each staff in your company according to Table 1 given. Make sure to show who will take the principle responsibility and who will take supportive responsibilities whenever necessary. Note that all staffs and all tasks listed in Table 1 should be in your LRC (3 marks)
- c. In addition, to implement this project, you need to choose a certain organisational structure (i.e, either functional, project based or matrix organisation). Justify your decision by describing how the organisational structure you have chosen will improve your ability to manage this project. You may need to make some assumptions. (3 Marks)

**Q2. AON Analysis (total 10 marks):** Complete a network analysis using AON approach to identify the critical path. Determine the early start, early finish, late start, late finish, total slack and free slack for each task. Use the following format of the table at each node. (Network model 2 marks, all tables in each node 7 marks and critical path 1 marks).  
(Answer: The project duration will be 66 days.)

|    |    |    |
|----|----|----|
| ES | ID | EF |
| TS |    | FS |
| LS | L  | LF |

**Table 1 Project tasks**

| Code | Task                                   | Duration Days | Precedence Relationships | Labour Resources                                                                               |
|------|----------------------------------------|---------------|--------------------------|------------------------------------------------------------------------------------------------|
| A    | Architectural decisions                | 4             | -                        | Design (Computing), Design (Mechanical)                                                        |
| B    | Hardware specifications                | 6             | A                        | Design (Computing), Design (Mechanical), Development (Mechanical), Assembly (Mechanical)       |
| C    | Software specifications                | 7             | A                        | Design(Computing), Development (Computing), Assembly (Computing)                               |
| D    | Conveyor design                        | 8             | B                        | Design (Mechanical), Development (Mechanical)                                                  |
| E    | Hardware design                        | 6             | B                        | Design (Computing), Design (Mechanical), Development (Mechanical)                              |
| F    | Software design                        | 12            | C                        | Design (Computing), Development (Computing)                                                    |
| G    | Operating system documentation         | 10            | C                        | Development (Computing), Documentation                                                         |
| H    | Hardware detail drawings               | 8             | E,F                      | Development(Computing), Development (Mechanical), Documentation                                |
| I    | Software programming                   | 16            | G                        | Design(Computing), Documentation                                                               |
| J    | Software verification/testing          | 12            | I                        | Development(Computing), Assembly(Computing), Documentation                                     |
| K    | Conveyor detailed drawings             | 7             | D                        | Development(Computing), Development (Mechanical), Documentation                                |
| L    | Drawing verification/Minor integration | 9             | H,K                      | Development (Computing), Assembly (Computing), Assembly (Mechanical)                           |
| M    | Prototype development                  | 4             | H                        | Development (Computing), Development (Mechanical)                                              |
| N    | Prototype installation                 | 7             | J,M                      | Assembly (Computing), Assembly (Mechanical),                                                   |
| O    | Hardware order/delivery                | 7             | L                        | Development (Mechanical), Purchase                                                             |
| P    | System Interface                       | 5             | L                        | Development (Mechanical), Assembly (Mechanical)                                                |
| Q    | Hardware assembly                      | 4             | O,P                      | Assembly (Computing), Assembly (Mechanical)                                                    |
| R    | Hardware/software Integration          | 5             | N,Q                      | Development (Computing), Development (Mechanical), Assembly (Computing), Assembly (Mechanical) |
| S    | Hardware/software documentation        | 2             | R                        | Development (Computing), Development (Mechanical), Documentation                               |
| T    | System verification                    | 3             | R                        | Development (Mechanical), Assembly (Computing), Assembly (Mechanical)                          |
| U    | Acceptance test                        | 2             | S,T                      | Assembly (Computing), Assembly (Mechanical)                                                    |

**Q3. Gantt chart with Resource Allocation Profile (total 10 marks):**

- (a) Draw an early start and a late start Gantt charts assuming that your company has an unlimited number of staff based on a work-pattern of 5 days per week (Monday to Friday), 8 hours per day and there are no public holidays (4 marks). (Answer: the duration should be 66 days).
- (b) Complete the resource allocation profile for each technical expert by identifying each staff resource requirement (i.e., for each Design (computing), Design (Mechanical), Development (Computing, Development (Mechanical), Assembly (computing), Assembly (Mechanical), and Documentation separately according to Table 1) throughout the duration of the project (3 marks)
- (c) Generate a labour cost chart in your early start Gantt chart to show the cost for each activity and the total cost for the whole project based on the cost for company staff provided in Table 2. Note that since the cost for the late start will be the same as the one in the early start, you do not need to calculate the cost chart for the late start Gantt chart. Do it only in the early start Gantt chart. Assume that we do not include the material cost in the cost calculation. (Answer: Total cost \$49,510) (3 marks)

**Q4. Slack management (5 marks):** Now, you have a constraint over the number of staff in your company. Assume that the number of staff in your company for each of expertise is as shown in Table 2. Note that there are two staffs in Development (Computing) and Documentation. Except those, in all the other expertise, we have only 1 staff in your company for each expertise. You can see that according to the resource allocation profile you have developed in Question 3, you have shortage problems of the resources (staff) because in some periods, your schedule requires more than 1 or 2 staffs that you have in your company. To solve this problem, the first step you have to take is to smooth the resource profile to reduce the peak demand. Now generate a new Gantt chart to smooth the resource demands (3 marks). Explain each adjustment you have made clearly to obtain the smoothed schedule (2 marks).

**Table 2 Cost for the company staff**

| Staffing Category        | Cost for Company Staff | Number of company staff |
|--------------------------|------------------------|-------------------------|
| Design (Computing)       | \$200/day              | 1                       |
| Design (Mechanical)      | \$200/day              | 1                       |
| Development (Computing)  | \$150/day              | 2                       |
| Development (Mechanical) | \$150/day              | 1                       |
| Assembly (Computing)     | \$100/day              | 1                       |
| Assembly (Mechanical)    | \$100/day              | 1                       |
| Purchase                 | \$80/day               | 1                       |
| Documentation            | \$80/day               | 2                       |

**Q5. Resource constraint problem (8 marks):** As you can see from the resource profile of the Gantt chart obtained from the smoothing process (in Question 4), you still have resource shortage problems if you use only the company staff. To solve these shortage problems, you can choose either to extend the duration of the project or to use other resources (like contractors and/or overtime during the weekends). In this question, let us assume that you use the first approach, to extend the project duration. This should be based on 5-days per week, working 8 hours per day, with no provision for overtime (no weekend work allowed). Also assume that once an activity is started, it cannot be interrupted.

- THE NUMBER OF STAFF IN EACH EXPERTISE IS GIVEN IN Table 2.
- All under graduate students (MFET 3008) should choose question (a), while all post graduate students (MFET 5040) should choose (b).
- Marking scheme: the answer itself (5 marks) & explanation and justification (3 marks).

(a)(For under graduate students (MFET 3008)): Using priority rule 1, determine the new minimum schedule for the project (Answer: 96 days):  
Priority rule 1=CPT-LS(i)

(b)(For post graduate students (MFET 5040)): Using a combination of priority rule 1 and priority rule 2 as given in the below, determine the new schedule for the project. The rule is determined as follows. For each activity  $i$ , you have to obtain,

- $PR(i)$ =Priority rule 1 obtained in (a).
- $Sum\_PR = \sum PR(i)$ = addition of all  $PR(i)$
- $Nor\_PR(i) = PR(i)/Sum\_PR$ =normalised priority rule 1.
- $PT(i)$ =The cost for each activity obtained in Question 3(c) which you have found in the early start Gantt chart (basically this is priority rule 2)
- $Sum\_PT = \sum PT(i)$ = addition of all  $PT(i)$
- $Nor\_PT(i) = PT(i)/Sum\_PT$ =normalised priority rule 2.

Once you find these, your priority rule will be given by

$$\text{Modified Priority Rule}(i) = [ Nor\_PR(i) \times 0.4 + Nor\_PT(i) \times 0.6 ] \times 1000.$$

So this basically puts a higher weight for the priority rule 2 (60%) than the priority rule 1 (40%).

The modified priority rule should be a full number only (i.e., do not use any decimal point numbers). You will see that the priority for A is 59, for B it is 78, and for C it is 75, etc.

Using this modified priority rule, determine a new schedule. (Answer: 96 days)

**Q6. Outsourcing and crashing (total 32 marks):** Now you need to solve the shortage of staff using the second approach, using contractors. To do it, you should use a systematic way so that all members in your company are happy with it. **Assume that you have only 1 staff in each expertise in this time (Use Table 4 – Do not use Table 2).**

(a) **Contract out (8 marks):** Firstly, you need to find the least expensive way to contract out the staffs having shortage problems. Starting from the schedule obtained in the slack management, determine which activities and which staff should be contracted to minimise the cost. You need to start from the schedule obtained in the smoothing process (Q4). However you need to modify the original schedule for your need. The work patterns and the rules to determine the contractors are given in Table 3. The cost for the company staff and a contract staff per day is shown in Table 4.

In the process, at each time you have contracted out a staff, you have to show a comparison chart for each different case of possibilities i.e., show all possible options which activities use contractors or which activities use company staffs and find the least expensive option from those options. See the following example.

Development (Mechanical) Staff is required by both activities S and T at day 62. You have two choices; you can contract out either S or T. The comparison will be

Option 1: Company staff for S and a contractor for T (T should include 3 days of induction)

Total cost:  $2(S) \times 150 + (3+3(T)) \times 250 = 1800$

Option 2: Company staff for T and a contractor for S (S should include 3 days of induction)

Total cost:  $3(T) \times 150 + (3+2(S)) \times 250 = 1700$

Option 2 is cheaper hence S should be contracted out.

**Table 3 Work Patterns and the rules for contracts**

1. The company normally works 5 days per week (Monday to Friday), 8 hours per day and there are no public holidays.
2. **Assume that the company has only 1 staff in each expertise as shown in Table 4. Do not use Table 2.**
3. If any activities can be started and the required staff is available, you have to start that activity as soon as possible.
4. For any activities, if your company staffs are available, you have to use him/her as much as possible.
5. The number of staff working on a task must remain constant over the duration of the task.
6. Once an activity is started it should be completed without any interruptions.
7. If a contract or company staff is used in an activity, he/she should complete that activity, i.e., no changes are allowed from company staff to contractor in the middle of activity, vice versa.
8. Where a contract staff is hired to overcome resource shortfalls, the contract staff

must be hired **3 working days earlier than required** to allow for the company induction. This must occur each time that a contract staff is employed as you will not be able to guarantee that he/she can conduct the contracted task on his/her first day of the work (meaning that for each contract staff used, you need to allow him/her to have 3 days of induction period regardless of that he/she has already worked for you).

9. This 3 days of induction will be applied for only the contract staff, i.e., if an activity needs two staffs (say A and B) and you decide to use a contract for only one of the two (say A), then the induction period will be applied for only contracted staff (A). The company staff (in the case B) can start from the original schedule date (no induction is required for the company staff).

**Table 4 Cost for the company and contract staff**

| Staffing Category        | Cost for Company Staff | Number of company staff | Cost for Contractors |
|--------------------------|------------------------|-------------------------|----------------------|
| Design (Computing)       | \$200/day              | 1                       | \$350/day            |
| Design (Mechanical)      | \$200/day              | 1                       | \$350/day            |
| Development (Computing)  | \$150/day              | 1                       | \$250/day            |
| Development (Mechanical) | \$150/day              | 1                       | \$250/day            |
| Assembly (Computing)     | \$100/day              | 1                       | \$200/day            |
| Assembly (Mechanical)    | \$100/day              | 1                       | \$200/day            |
| Purchase                 | \$80/day               | 1                       | \$150/day            |
| Documentation            | \$80/day               | 1                       | \$150/day            |

You have to calculate the direct cost of the whole project again including all contractors. Make sure you have included for 3 days induction for all contract staff. Since the cost for contractors is higher than the cost for staff (as shown in Table 4), the cost will be increased compared to the answer you have obtained in Question 3 – Early Start Gantt chart. Please fully explain how you obtain your answer. (Answer: Some (or all) staff in activities B, E, F, I, J, L, M, N, P, Q and S are contracted out. In this case, the schedule with all contractors would cost \$70,230 with 66 days working only weekdays. Note that your answer should be at least this much. If your answer is lower than this figure and the answer is “CORRECT”, you will have 20% bonus mark on this question.)

- (b) **Crashing (24 marks):** As you can see, the total number of days for the normal duration is 66 days (which is 13 weeks and 1 day, hence you have to consider it as 14 weeks). Now your task is to reduce the number of weeks to complete this project using overtime on Saturdays and Sundays. Based on labour costs alone (as material costs are common to all schedules), determine the optimum project length which provides the least cost at the completion using the rules listed in Table 5. At every week of crashing, you should generate a separate time-cost chart (See the MSEXcel template provided), which clearly shows how you made the crashing. Clearly show how all costs have been calculated in each crashing. Full explanation for your decision is critical since that is your justification for your decision.

- The crashing should be made by working overtime during weekends. The overtime expense should be paid to **both the company staff and the contractors**. The additional payment due to working overtime will be the crashing cost for each task. The crashing should be made in week by week basis. Overtime can only be worked on Saturdays and Sundays and this is for only eight hours maximum on each day. You cannot increase the hours worked in any day.  
Overtime expenses:     Saturdays: 1.5 times normal labour cost  
                              Sundays: 3 times normal labour cost
- For the convenience of calculation, simply assume that all Saturdays should be used before any Sundays are used for overtime.**
- For the crashing, your priority rule is always to select any activities that require the least crashing cost than the others.**
- If the cost for crashing is the same for several different options, then you should choose the earliest one. For example, to use Saturday of week 2 and Saturday of week3 cost exactly the same, then you should use Saturday of week 2: earlier one!**
- For the convenience of the calculation, once a decision is made on which activities are contracted out at the normal schedule (which is in Q6(a)), you have to stick to this decision for the further crashing. For example, in the normal schedule, if you decide to contract out Design(Computing) for activity B and Assembly(Computing) for L, then for the further crashing, you should use these as contracted, i.e., you do not need to re-evaluate which activities should be contracted out for every crashing case. The contracted staffs chosen in the normal schedule should be used to the end of the crashing.
- Note that when it is crashed, the last day of the last week should be Saturday. This means that you cannot use Sunday of the last week. So for example, for the 12 weeks of schedule, the last day in week 12 should be Saturday of week 12. Also this means that for 13 weeks of schedule, you should use activity U for crashing.
- Hint: Firstly you have to crash necessary critical activities. Some non-critical activities could become critical during the smoothing process or during the crashing. Hence some non-critical activities also need to be crashed. So every time you crash, check whether non-critical activities become critical.

In the process, at each crashing, you have to consider all possible overtime options in each week, and then you need to choose the least expensive ones. As you can see that in the first crashing (from 14 weeks to 13 weeks), you should use “U” since the last day of the last week is Saturday. So we need to overtime U to make the last day of week 13 Saturday (the total direct cost will be \$70,330). However, for the further processing, you have to show all possible crashing options and choose the least expensive one. It is important that you have to show all this processes in your submission. See the following example for the second crashing (from 13 weeks to 12 weeks). Note that this is an example so this is not necessarily to be correct. From the



Gantt chart you generated in Week 13 schedule, you can identify which activities should be crashed on Saturday in each week. For example, in week 1, activity C should be crashed but you do not need to crash B due to the slacks B has. You can calculate the cost for crashing as well (as 0.5 times of the total cost per each day for each activity). The result can be shown like a table below. As you can see that Week 2 and Week 3 cost least, you should choose week 2 and 3 for crashing. You have to complete this table to find additional 3 more Saturdays to make the schedule 12 weeks. (Answer: The total direct cost for 13 week schedule is then \$70,345). You have to do this for each crashing.

| Week | Description                                                   | Cost                                          |
|------|---------------------------------------------------------------|-----------------------------------------------|
| 1    | C: to be crashed.<br>B: no need (slacks)                      | $450(C) \times 0.5 = 225$                     |
| 2    | G: to be crashed.                                             | $230(G) \times 0.5 = 115$                     |
| 3    | G: to be crashed<br>D, E & F: no need (slacks)                | $230(G) \times 0.5 = 115$                     |
| 4    | I: to be crashed<br>K: Also need to crash. F: no need (Slack) | $350(I) \times 0.5 + 380(K) \times 0.5 = 365$ |
| 5    |                                                               |                                               |
| 6    |                                                               |                                               |
| 7    |                                                               |                                               |
| 8    |                                                               |                                               |
| 9    |                                                               |                                               |
| 10   |                                                               |                                               |
| 11   |                                                               |                                               |
| 12   | U crashed (Already) used                                      |                                               |

You need to consider both “direct” cost (cost for crashing and labour cost) and “indirect” cost (O/H cost, bonus and penalty) to find the optimum schedule which provides you the one with the minimum total cost. The bonus, penalties and the overhead cost are given in Table 6.

**Table 6 Penalties and Bonuses for crashing**

1. The project has a contractual completion date at the end of week 12 and the penalties are applied for any later completion. The penalty has been established at \$2000 for every week or part of the week that the project extends beyond week 12 (meaning that no penalty is applied if the project is finished at week 12.) The penalty is started from week 13 (with \$2000 penalty) and in the case it is completed at week 14, the penalty will be \$2000 x 2.
2. A bonus payment is available, if the project is finished before week 12. The

bonus is \$3000 for each week that the project is completed before week 12 (meaning that there is \$3000 bonus if the project is completed at week 11. It will be \$3000 x 2 if the project is finished at week 10.)

3. The project involves an overhead cost. The overhead cost of the project is \$500 per week and this must be charged to the project for every week (or part of) that the project is in progress.

**Q7. EV Analysis (total 15 marks):** Do the Earned value analysis. Note that you might need to assume that some activities use **start-to-start precedence** relationships rather than finish-to-start to cope with delays in the schedule. If that is the case, clearly state it in your answer.

- (a) Complete an Earned Value analysis based on the optimum schedule obtained in Q6(b) and the project control data given in Table 7 at the end of week 5. The expenditure values shown in the table are those spent on labour only (9 marks).
- (b) Draw a Gantt chart that shows planned and actual progress (2 marks).
- (c) Determine the expected budget at completion and the expected completion date for the project using both the original and revised approaches. Explain the assumptions that have been made in the determination of your estimations (4 marks).

**Table 7 Project Control data at the end of week 5**

| Code | Current Expenditure | Percentage Complete |
|------|---------------------|---------------------|
| A    | \$ 2,400            | 100                 |
| B    | \$ 6,500            | 100                 |
| C    | \$ 4,300            | 100                 |
| D    | \$ 3,500            | 100                 |
| E    | \$ 6,100            | 100                 |
| F    | \$ 8,500            | 50                  |
| G    | \$ 2,100            | 100                 |
| H    | \$2,000             | 10                  |
| I    | \$5,300             | 20                  |
| K    | \$3,200             | 80                  |

**Q8. Professional presentation (10 marks):** When assessing the report, 10 Marks will be allocated for the professionalism and clarity of the submission.

## Marking sheet for case Study

### Project planning and Control (G)

| Q          | Component                                               | Mark Avail. | Mark Given | Comment |
|------------|---------------------------------------------------------|-------------|------------|---------|
| 1          | OBS                                                     | 2           |            |         |
|            | WBS                                                     | 2           |            |         |
|            | LRC                                                     | 3           |            |         |
|            | Organisational structure selection                      | 3           |            |         |
| 2          | AON Network                                             | 2           |            |         |
|            | Table at each node                                      | 7           |            |         |
|            | Critical Path                                           | 1           |            |         |
| 3          | Early Start                                             | 2           |            |         |
|            | Late Start                                              | 2           |            |         |
|            | Resource Allocation Profile                             | 3           |            |         |
|            | Labour Cost chart for Early Start                       | 3           |            |         |
| 4          | Slack management                                        | 3           |            |         |
|            | Explanation/Justification                               | 2           |            |         |
| 5          | Resource Constraint Schedule                            | 5           |            |         |
|            | Explanation/Justification                               | 3           |            |         |
| 6          | Normal schedule with contractors/explanation            | 8           |            |         |
|            | Crash 1 schedule/explanation                            | 2           |            |         |
|            | Crash 2 schedule/explanation                            | 8           |            |         |
|            | Crash 3 schedule/explanation                            | 4           |            |         |
|            | Crash 4 schedule/explanation                            | 8           |            |         |
|            | Optimum Selection process                               | 2           |            |         |
| 7          | Earned value analysis                                   | 5           |            |         |
|            | Progress Report for each activity and the whole project | 4           |            |         |
|            | Gantt chart showing the progress                        | 2           |            |         |
|            | Expected budget at completion                           | 2           |            |         |
|            | Expected Time of completion                             | 2           |            |         |
| 8          | Quality of submission                                   | 10          |            |         |
| total mark |                                                         | 100         |            |         |